

Poisson Extremes and Gaussian Bridges under Small-Sum Conditioning

Localization, incomplete-gamma laws, and
all-order logarithmic corrections

Prepared for Vladimir Reshetnikov
30 September 2026

Abstract

For a positive summable weight sequence, let $S = \sum_j w_j U_j$, with independent uniform U_j on $[0, 1]$, and condition on $S \leq x$ as $x \downarrow 0$. We investigate the locations and heights of the largest summands, jointly with the fluctuation process of the whole configuration. For $w_j = j^{-p}$, $p > 1$, write $N = t^{1/p}$ for the effective dimension at the mean-matching exponential tilt t , put $\beta = 1/p$, and define $b + \beta \log b = \log N$. We prove, uniformly for bounded z ,

$$\log \mathbb{P} \left(\max_j t j^{-p} U_j \leq b + z \mid S \leq x \right) = -N \Gamma(1 - \beta, b + z) + O_p(N^{-1/2}).$$

Consequently every fixed-order logarithmic correction is explicit. The extremal point process, with locations divided by N/b^β , tends to a Poisson process of intensity $\mathbf{1}_{[0,1]}(u) du e^{-z} dz$. It is jointly independent in the limit of the variance-clock Brownian bridge and the exponential distance from the conditioning boundary.

For every uniformly lacunary weight sequence, we also prove the full marked Poisson limit and its joint independence from the bridge and the boundary distance. This settles the qualitative point-process and independence portions of Question 7 in the repository's lacunary-conditioning report; that question's quantitative-rate requirement remains open. The polynomial extreme-location scale and the all-order maximum law are additional extensions. Proofs, reproducible symbolic checks, and numerical Fourier-inversion diagnostics are supplied. No Lean certification, external refereeing, or exhaustive worldwide priority determination is claimed.

Inspected repository snapshot. a4268e78ebd0f6bf71ba609c07f4b3415748f532.

Research status. Complete manuscript proofs of the displayed results; candidate contributions relative to the inspected sources.

Contents

1	Research target and the main results	1
2	Exponential tilting and the effective dimension	3
3	A quantitative local limit theorem and conditioning identities	5
4	The incomplete-gamma law for the maximum	7
5	Poisson localization for polynomial weights	9
6	The Gaussian bridge and joint independence	10
7	Completion of the qualitative lacunary point-process question	13
8	Reproducible symbolic and numerical diagnostics	14
9	Proof status, formalization, and limitations	16
10	Further research questions	17
11	Conclusion	18
A	Reproduction and source provenance	18

1 Research target and the main results

1.1 What the repository already establishes

The Fabius–Rvachev part of ProveIt develops the probability distribution associated with $\sum_{j \geq 1} 2^{-j} U_j$. This probabilistic interpretation is classical; it appears, with the corresponding shift of normalization, in Arias de Reyna’s Theorem 3 [1]. Our polynomially weighted series is a related model, not another normalization of the Fabius function.

Two inspected repository manuscripts are particularly relevant. The lacunary report [5] studies full-configuration approximations, variance-fraction effects, a Brownian bridge, an exponential boundary slack, and a Gumbel maximum. Its Question 7 asks for the entire extremal point process and for joint independence from the bridge, with explicit rates. The report expressly does not claim that independence. A second manuscript [6] already gives a general variance-fraction conditioning theory for arbitrary positive summable uniform weights, including polynomial variance clocks. Thus extending that variance criterion to j^{-p} is *not* a new result of the present article.

The contributions developed here concern a more refined question: *where do the extremes occur, how accurately can their distribution be calculated, and are they independent of the macroscopic bridge?* Theorem 1.3 answers the qualitative parts of the repository’s Question 7 throughout its uniformly lacunary class. Theorems 1.1 and 1.2 supply the distinct polynomial-weight theory, including a quantitative maximum law strong enough to imply every fixed-order inverse-logarithmic correction.

Exponential tilting and increasing-block conditional approximation are classical, not originality claims: see Diaconis–Freedman [2] and Dembo–Zeitouni [3]. Poisson limits for independent rare events and Gaussian bridge conditioning are likewise standard mechanisms. We give their needed arguments explicitly. The proposed addition is the theorem package for the specified infinite, increasingly tilted arrays, the exact incomplete-gamma reduction, and the joint limits. The literature review was targeted, rather than exhaustive. None of these statements is identified here with the resolution of a named published conjecture outside the repository.

1.2 Notation and the conditioning convention

All uniforms are independent under the original probability measure P . For a positive summable sequence (w_j) , put

$$S = \sum_{j \geq 1} w_j U_j, \quad L(t) = \mathbb{E}_P e^{-tS}, \quad \frac{dQ_t}{dP} = \frac{e^{-tS}}{L(t)}. \quad (1.1)$$

Choose $t > 0$ so that $x = \mathbb{E}_{Q_t} S$, and write

$$a_j = tw_j, \quad Y_j = a_j U_j, \quad T = \sum_j Y_j, \quad \mu = \mathbb{E}_{Q_t} T = tx, \quad V = \text{Var}_{Q_t} T, \quad P_x = P(\cdot \mid S \leq x). \quad (1.2)$$

The nonnegative slack under P_x is $H = \mu - T = t(x - S)$. All conditioning in the main results is on the positive-probability inequality $S \leq x$, not on a null event $S = x$. The parameter t tends to infinity as x tends to zero.

Let $m_j = \mathbb{E}_{Q_t} Y_j$ and $v_j = \text{Var}_{Q_t} Y_j$. Define the exact variance clock and its polygonal process by

$$r_0 = 0, \quad r_k = V^{-1} \sum_{j \leq k} v_j, \quad \mathcal{B}_t(r_k) = V^{-1/2} \sum_{j \leq k} (Y_j - m_j). \quad (1.3)$$

Linearly interpolate between consecutive r_k , and set $\mathcal{B}_t(1) = (T - \mu)/\sqrt{V}$. This gives a continuous function on $[0, 1]$: both $\sum Y_j$ and $\sum m_j$ are finite. A standard Brownian bridge is denoted B° ; its covariance is $\min(r, s) - rs$. The standard normal density is φ .

Point processes below are random locally finite counting measures on $[0, \infty) \times \mathbb{R}$, with the vague topology. A Poisson point process with intensity $\mathbf{1}_{[0,1]}(u) du e^{-z} dz$ is denoted Ξ . It has finitely many points above any fixed height and arbitrarily many points towards height $-\infty$. Its boundary lines at $u = 0, 1$ have zero intensity. Weak convergence of processes refers to the uniform topology on $C([0, 1])$; convergence of the triples uses the product topology.

1.3 Polynomial weights: quantitative extremes

For the rest of this subsection take $w_j = j^{-p}$, where $p > 1$ is fixed, and define

$$\beta = \frac{1}{p}, \quad N = t^\beta, \quad b + \beta \log b = \log N, \quad M = Nb^{-\beta} = e^b. \quad (1.4)$$

For $N > 1$ the equation for b has a unique positive solution. Neither N nor M needs to be an integer. Also

$$b = \log N - \beta \log \log N + o(1), \quad M \sim \frac{N}{(\log N)^\beta}. \quad (1.5)$$

We use the upper incomplete gamma function $\Gamma(s, y) = \int_y^\infty e^{-v} v^{s-1} dv$.

Theorem 1.1 (Incomplete-gamma maximum law). *For every fixed $p > 1$ and $Z < \infty$, as $N \rightarrow \infty$,*

$$\sup_{|z| \leq Z} \left| \log P_x \left(\max_{j \geq 1} Y_j \leq b + z \right) + N\Gamma(1 - \beta, b + z) \right| = O_{p,Z}(N^{-1/2}). \quad (1.6)$$

More precisely, the proof gives an error $O_{p,Z}(N^{-1/2} + b^2/N + e^{-b})$; the last two terms are $o(N^{-1/2})$. For every fixed integer $K \geq 0$,

$$\log P_x \left(\max_j Y_j \leq b + z \right) = -e^{-z} \sum_{r=0}^K \frac{(-1)^r (\beta)_r}{b^r} \sum_{k=0}^r \frac{z^k}{k!} + O_{p,Z,K}(b^{-K-1} + N^{-1/2}), \quad (1.7)$$

uniformly for $|z| \leq Z$, where $(\beta)_0 = 1$ and $(\beta)_r = \beta(\beta + 1) \cdots (\beta + r - 1)$.

In particular, the first three terms in the brackets are

$$1 - \frac{\beta(z+1)}{b} + \frac{\beta(\beta+1)(z^2/2 + z + 1)}{b^2}. \quad (1.8)$$

The logarithmic expansion is asymptotic order by order. It is not claimed to converge, to have a uniform remainder for growing K , or to remain valid when z is unbounded. The theorem implies the Gumbel law $P_x(\max_j Y_j - b \leq z) \rightarrow \exp(-e^{-z})$.

Theorem 1.2 (Locations, bridge, and slack for polynomial weights). *Let*

$$\Xi_t = \sum_{j \geq 1} \delta_{(j/M, Y_j - b)}. \quad (1.9)$$

Under P_x ,

$$(\mathcal{B}_t, \Xi_t, H) \Rightarrow (B^\circ, \Xi, E), \quad E \sim \text{Exp}(1), \quad (1.10)$$

and the three limiting objects are mutually independent. If J_t is the almost surely unique index attaining $\max_j Y_j$, then

$$\left(\frac{J_t}{M}, \max_j Y_j - b \right) \Rightarrow (U, G), \quad (1.11)$$

where U is uniform on $[0, 1]$, $P(G \leq z) = e^{-e^{-z}}$, and U, G are independent. This pair is also independent of the limiting bridge and slack.

The localization is genuinely submacroscopic: $M/N \rightarrow 0$, whereas J_t/M has a nondegenerate limit. A variance clock alone does not reveal this scale. The upper caps $a_j = (N/j)^p$ force it.

1.4 Lacunary weights: the repository's qualitative question

Suppose instead that

$$w_j > 0, \quad w_{j+1} \leq q w_j \quad (j \geq 1), \quad 0 < q < 1, \quad n = \#\{j : t w_j \geq 1\}. \quad (1.12)$$

Set $\Xi_t^{\text{lac}} = \sum_j \delta_{(j/n, Y_j - \log n)}$.

Theorem 1.3 (Marked extremes for every uniformly lacunary sequence). *Under (1.12), as $x \downarrow 0$,*

$$(\mathcal{B}_t, \Xi_t^{\text{lac}}, H) \Rightarrow (B^\circ, \Xi, E) \quad (1.13)$$

under P_x , with the same intensity and mutual independence as in Theorem 1.2. In particular the limiting location of the maximum, on the scale n , is uniform and independent of its Gumbel height, the bridge, and the exponential slack.

This includes $w_j = 2^{-j}$ and arbitrary supergeometric gaps compatible with (1.12). It proves the full point-process convergence and joint independence asked for in Question 7 of [5]. No quantitative rate in a metric on the joint path–point-process space is supplied; that portion of the question remains a research problem.

2 Exponential tilting and the effective dimension

2.1 Existence, independence, and exact densities

The sum S lies in $[0, \sum w_j]$. Every sufficiently small $x > 0$ has $P(S \leq x) > 0$: first make the deterministic tail less than $x/2$, and then restrict finitely many uniforms so their weighted sum is below $x/2$. Differentiation under the bounded integral gives

$$-\frac{d}{dt} \log L(t) = \mathbb{E}_{Q_t} S, \quad \frac{d}{dt} \mathbb{E}_{Q_t} S = -\text{Var}_{Q_t} S < 0.$$

The tilted mean is continuous, starts at $\frac{1}{2} \sum w_j$, and tends to zero. The last assertion follows by dominated convergence for each coordinate and the deterministic summable tail bound. Hence the mean-matching t exists and is unique for $0 < x < \frac{1}{2} \sum w_j$.

Finite-dimensional factorization and dominated convergence show that under Q_t the Y_j are independent and have densities

$$p_a(y) = \frac{e^{-y}}{1 - e^{-a}} \mathbf{1}_{(0,a)}(y), \quad a = a_j. \quad (2.1)$$

Integration or differentiation of the normalizing factor gives

$$m(a) = 1 - \frac{a}{e^a - 1}, \quad v(a) = 1 - \frac{a^2 e^a}{(e^a - 1)^2}, \quad v(a) = m(a) - a m'(a). \quad (2.2)$$

In particular $0 < v(a) \leq 1$. Near zero,

$$m(a) = a/2 - a^2/12 + O(a^4), \quad v(a) = a^2/12 + O(a^4),$$

and as $a \rightarrow \infty$ the laws in (2.1) tend to $\text{Exp}(1)$.

Lemma 2.1 (Uniform moments). *For every integer $r \geq 2$ there are positive constants depending only on r such that, for $Y \sim p_a$ and $X = Y - m(a)$,*

$$c \min(a^2, 1) \leq v(a) \leq C \min(a^2, 1), \quad \mathbb{E}|X|^r \leq C_r \min(a^r, 1), \quad \mathbb{E}X^4 \leq C v(a)^2. \quad (2.3)$$

Proof. For $a \leq 1$, $|X| \leq a$, and $v(a)/a^2$ extends continuously to $1/12$ at zero and stays positive. For $a \geq 1$, the density is bounded above by $(1 - e^{-1})^{-1} e^{-y}$, while $m(a) \leq 1$, giving all uniform centered moment bounds. The variance has a positive minimum on $[1, \infty)$, since it is continuous, positive, and tends to one. Combining these facts proves the final bound as well. $\square$

2.2 Polynomial constants and the physical endpoint

Proposition 2.2 (The polynomial variance clock). *For $a_j = (N/j)^p$, $p > 1$, define*

$$A_p = \int_0^\infty m(u^{-p}) du, \quad B_p = \int_0^\infty v(u^{-p}) du, \quad (2.4)$$

$$C_p = \int_0^\infty \log \frac{u^{-p}}{1 - e^{-u^{-p}}} du. \quad (2.5)$$

These are finite and positive, and

$$\mu \sim A_p N, \quad V \sim B_p N, \quad -\log L(t) \sim C_p N, \quad (2.6)$$

$$A_p = \beta C_p, \quad B_p = (1 - \beta) A_p, \quad C_p = -\Gamma(1 - \beta) \zeta(1 - \beta). \quad (2.7)$$

For every $r \geq 2$, $\sum_j \mathbb{E}_{Q_t} |Y_j - m_j|^r = O_{p,r}(N)$. The index clock converges to

$$F_p(u) = \frac{1}{B_p} \int_0^u v(s^{-p}) ds, \quad r_{\lfloor Nu \rfloor} \rightarrow F_p(u), \quad 0 \leq u < \infty. \quad (2.8)$$

Moreover

$$x \sim A_p t^{\beta-1}, \quad t \sim (A_p/x)^{1/(1-\beta)}, \quad N \sim (A_p/x)^{1/(p-1)}. \quad (2.9)$$

Proof. Riemann sums apply after separately controlling $j < \delta N$ and $j > RN$. For m and v , the small-index summands are at most one; the large-index tails are bounded respectively by constants times $(N/j)^p$ and $(N/j)^{2p}$. For the logarithmic integrand the behavior at $u = 0$ is $-p \log u + o(1)$ and at infinity it is $u^{-p}/2 + O(u^{-2p})$. The endpoint contributions are integrable and uniformly negligible as $\delta \downarrow 0$ and $R \uparrow \infty$. This proves the three asymptotics and the clock limit. Lemma 2.1 proves the moment sum by the same head-tail split.

Put $g(a) = \log[a/(1 - e^{-a})]$. Then $g'(a) = m(a)/a$. The substitution $a = u^{-p}$ and integration by parts give

$$C_p = \beta \int_0^\infty g(a) a^{-\beta-1} da = \int_0^\infty m(a) a^{-\beta-1} da = A_p/\beta.$$

All boundary terms vanish: $g(a) = O(a)$ at zero and $g(a) = O(\log a)$ at infinity. Similarly, using (2.2),

$$B_p = \beta \int_0^\infty [m(a) - am'(a)] a^{-\beta-1} da = (1 - \beta) A_p.$$

For completeness, the regularized Euler integral for $0 < s < 1$ is

$$\Gamma(s) \zeta(s) = \int_0^\infty \left(\frac{1}{e^a - 1} - \frac{1}{a} \right) a^{s-1} da. \quad (2.10)$$

It follows from the usual Euler integral for $s > 1$ by splitting at one, subtracting $1/a$ on $(0, 1)$, and analytically continuing the explicit term $1/(s - 1)$; on $0 < s < 1$ that term equals the integral of $-1/a$ over $(1, \infty)$. The subtracted integral is absolutely convergent. Taking $s = 1 - \beta$ and using $m(a) = -a[(e^a - 1)^{-1} - a^{-1}]$ proves the last identity in (2.7). Standard gamma and zeta integral conventions are as in [4]. Finally $x = \mu/t$ proves (2.9). $\square$

For $p = 2$ the constants are

$$C_2 = 2.588410972826785 \dots, \quad A_2 = 1.294205486413393 \dots, \quad B_2 = 0.647102743206696 \dots$$

These scalar facts are background for the new extremal results, not a claim to supersede the polynomial variance analysis in [6].

2.3 The lacunary dimension

Proposition 2.3. *Under (1.12), $n \rightarrow \infty$ and*

$$\mu = n + O_q(1), \quad V = n + O_q(1), \quad \sum_j \mathbb{E}|Y_j - m_j|^r = O_{q,r}(n), \quad r \geq 2. \quad (2.11)$$

Uniformly in $k \geq 0$,

$$\sum_{j \leq k} v_j = \min(k, n) + O_q(1). \quad (2.12)$$

Proof. For $j \leq n$, $a_j \geq q^{-(n-j)}$. For $k \geq 1$, $a_{n+k} < q^{k-1}$. The deficits $1 - m(a)$ and $1 - v(a)$ on the active part are summable along the first bound, because they decay exponentially as $a \rightarrow \infty$; they can also be bounded uniformly for $a \geq 1$. The tails $m(a) = O(a)$ and $v(a) = O(a^2)$ are summable along the second bound. The same argument and Lemma 2.1 give the moment estimate and the uniform clock assertion. Every fixed coordinate becomes active as $t \rightarrow \infty$, proving $n \rightarrow \infty$. $\square$

3 A quantitative local limit theorem and conditioning identities

We give the analytic estimates used below instead of relying on an unproved transfer of an independent maximum law through a rare event. Write $R = N$ in the polynomial case and $R = n$ in the lacunary case. Constants may depend on the fixed parameter p or q .

Lemma 3.1 (A smoothing block). *For either array, every coordinate set I with $V_I := \sum_{j \in I} v_j \geq \varepsilon R$ contains at least $c_\varepsilon R$ coordinates whose caps are at least $a_\varepsilon > 0$. The same assertion holds, uniformly in $y \geq 2$, for the polynomial array with caps $c_j = \min(a_j, y)$ and with its own variances.*

Proof. In the polynomial case the first δN coordinates contribute at most $\delta N + 1$ variance. The part beyond AN contributes at most $C_p N A^{1-2p}$, by Lemma 2.1; here A is a fixed large constant. Choose δ small and A large so their combined contribution is less than $\varepsilon N/2$. At least $\varepsilon N/2$ variance remains in $[\delta N, AN]$, so at least $\varepsilon N/2$ coordinates remain there, since each variance is at most one. Their original caps are at least A^{-p} . Capping at $y \geq 2$ preserves a positive lower bound on those caps and cannot increase the two discarded variance bounds.

In the lacunary case the entire inactive tail has $O_q(1)$ variance. Thus I contains at least $\varepsilon n/2$ active coordinates for large n , and all their caps are at least one. $\square$

Lemma 3.2 (Uniform density local limit theorem). *Let I be any coordinate set with $V_I \geq \varepsilon R$. The sum $T_I = \sum_{j \in I} Y_j$ has a continuous density f_I , and*

$$\sup_{z \in \mathbb{R}} \left| \sqrt{V_I} f_I(\mu_I + \sqrt{V_I} z) - \varphi(z) \right| \leq C_\varepsilon R^{-1/2}, \quad \mu_I = \mathbb{E} T_I. \quad (3.1)$$

This is also uniform for the polynomial capped arrays of Lemma 3.1. In particular $\sup f_I \leq C_\varepsilon R^{-1/2}$.

Proof. For a centered variable with density (2.1), write $\chi_a(s)$ for its characteristic function. Apart from its unit-modulus centering factor, it is

$$\frac{1 - e^{-(1-is)a}}{(1 - e^{-a})(1 - is)} \quad (3.2)$$

Uniformly for $a \geq a_0 > 0$, this has modulus at most $C_{a_0}/|s|$ for $|s| \geq 1$. On any compact frequency interval $\eta \leq |s| \leq A$ it has modulus at most some $\rho < 1$, uniformly for $a \geq a_0$. To see strict uniformity, adjoin $a = \infty$: the expression converges there to $(1 - is)^{-1}$, while at every finite positive a a variable with a positive interval density has characteristic-function modulus strictly below one away from zero. Compactness then applies.

The moment bounds also imply, uniformly for sufficiently small $|s|$,

$$\log \chi_a(s) = -\frac{1}{2}v(a)s^2 + O\left(\min(a^3, 1)|s|^3\right). \quad (3.3)$$

For $a \leq 1$ this follows by Taylor expansion of a bounded centered variable, whose variance is comparable to a^2 ; for $a \geq 1$ the uniform exponential moment bound supplies the same expansion on a fixed neighborhood of zero. The logarithm is taken on its continuous branch equal to zero at $s = 0$.

Let $\Psi_I(u)$ be the characteristic function of $(T_I - \mu_I)/\sqrt{V_I}$. The sum of the error coefficients in (3.3) is $O(R)$, and $V_I \asymp_\varepsilon R$. For $|u| \leq R^{1/12}$,

$$\Psi_I(u) = e^{-u^2/2} \left[1 + O_\varepsilon(R^{-1/2}|u|^3) e^{O_\varepsilon(R^{-1/2}|u|^3)} \right].$$

Its integrated difference from $e^{-u^2/2}$ on that interval is $O_\varepsilon(R^{-1/2})$. Between $R^{1/12}$ and $\eta\sqrt{V_I}$, the small-frequency modulus bound from (3.3), with η chosen sufficiently small, gives $|\Psi_I(u)| \leq e^{-c_\varepsilon u^2}$. Between $\eta\sqrt{V_I}$ and $A\sqrt{V_I}$ the smoothing block gives $|\Psi_I(u)| \leq \rho^{c_\varepsilon R}$. Above $A\sqrt{V_I}$ it gives

$$|\Psi_I(u)| \leq \left(\frac{C\sqrt{V_I}}{|u|} \right)^{c_\varepsilon R}.$$

Choose $A > 2C$. Integrating these three bounds, and the analogous Gaussian tails, contributes an exponentially small quantity. Thus $\int_{\mathbb{R}} |\Psi_I(u) - e^{-u^2/2}| du = O_\varepsilon(R^{-1/2})$.

For large R the smoothing block has more than two coordinates, so its characteristic function is integrable. The remaining independent coordinates have characteristic-function modulus at most one. The same argument therefore applies to infinite I , by convergence of the sums and dominated Fourier inversion. The inversion inequality now proves (3.1) and the density bound. $\square$

3.1 The exact likelihood and the slack

Let f be the tilted density of T , and define

$$D = \mathbb{E}_{Q_t} [e^{T-\mu} \mathbf{1}_{T \leq \mu}] = \int_0^\infty e^{-h} f(\mu - h) dh. \quad (3.4)$$

Changing measure in (1.1) gives the exact identities

$$\frac{dP_x}{dQ_t} = D^{-1} e^{T-\mu} \mathbf{1}_{T \leq \mu}, \quad P(S \leq x) = L(t) e^\mu D. \quad (3.5)$$

It is important that this full-product likelihood is not close to one. Approximation of selected marginals requires integrating out the complement.

Proposition 3.3 (Normalization and boundary slack). *In either array,*

$$D = (2\pi V)^{-1/2} (1 + O(R^{-1/2})). \quad (3.6)$$

Under P_x , H tends in total variation to $\text{Exp}(1)$, and $\mathbb{E}_{P_x} H^r \rightarrow r!$ for every fixed nonnegative integer r . In the polynomial case,

$$-\log P(S \leq x) \sim (1 - \beta) C_p N. \quad (3.7)$$

Proof. Apply Lemma 3.2 to T in (3.4). After multiplying by $\sqrt{V}$, the integral of the error is $O(R^{-1/2})$ and

$$\int_0^\infty e^{-h} \varphi(h/\sqrt{V}) dh = (2\pi)^{-1/2} (1 + O(V^{-1})).$$

This proves (3.6). The density of H is exactly $e^{-h} f(\mu - h)/D$, $h \geq 0$. It tends pointwise to e^{-h} and is bounded by $C e^{-h}$, by the density bound in Lemma 3.2. Dominated convergence proves both total variation and the moments. Finally use (3.5), Proposition 2.2, and $\log D = O(\log N)$. $\square$

For $I \subseteq \mathbb{N}$ put $Z_I = T_I - \mu_I$ and $J = I^c$. When $V_J \geq \varepsilon R$, the joint law of $((Y_j)_{j \in I}, H)$ has density relative to $Q_t^I \otimes dh$ equal to

$$R_I(\mathbf{y}, h) = e^{-h} \frac{f_J(\mu_J - Z_I(\mathbf{y}) - h)}{D} \mathbf{1}_{h \geq 0}. \quad (3.8)$$

This follows by disintegration of the independent sums; it applies to infinite coordinate sets because their nonnegative sums are measurable limits. In particular $R_I(\mathbf{y}, h) \leq C_\varepsilon e^{-h}$.

Corollary 3.4 (Negligible blocks with independent slack). *If $V_I = o(R)$, then*

$$d_{\text{TV}}\left(\mathcal{L}_{P_x}((Y_j)_{j \in I}, H), Q_t^I \otimes \text{Exp}(1)\right) \longrightarrow 0. \quad (3.9)$$

More generally, if $V_{I^c} \geq \varepsilon R$, the marginal likelihood of P_x^I with respect to Q_t^I is bounded by C_ε .

Proof. The general bound follows by integrating (3.8) in h . In the negligible-block case, $Z_I/\sqrt{R} \rightarrow 0$ in Q_t -probability by Chebyshev's inequality, and $V_{I^c}/V \rightarrow 1$. Lemma 3.2 and (3.6) show that the ratio in (3.8) tends to one in probability for each fixed h . Its uniform bound, followed by integration against $e^{-h} dh$, gives convergence in $L^1(Q_t^I \otimes \text{Exp}(1))$. Half this L^1 distance is total variation. $\square$

This special case of the general variance principle is reproduced for the proofs below; the broader principle is already in [6].

4 The incomplete-gamma law for the maximum

The proof separates an exact independent-array calculation from a quantitative conditioning estimate. Merely knowing that an extreme prefix has negligible variance would give an $o(1)$ error, insufficient for every logarithmic correction.

4.1 An exact intensity integral

For $y > 0$, the tilted exceedance probability at coordinate j is

$$p_j(y) = \frac{e^{-y} - e^{-a_j}}{1 - e^{-a_j}} \mathbf{1}_{a_j > y}. \quad (4.1)$$

Only finitely many coordinates can exceed y . Put $q_j(y) = (e^{-y} - e^{-a_j})_+$ and $\Lambda_N(y) = \sum_j p_j(y)$.

Lemma 4.1 (Integral and lattice errors). *For $a_j = (N/j)^P$,*

$$\int_0^\infty (e^{-y} - e^{-(N/s)^P})_+ ds = N\Gamma(1 - \beta, y). \quad (4.2)$$

Furthermore

$$\left| \sum_{j \geq 1} q_j(y) - N\Gamma(1 - \beta, y) \right| \leq e^{-y}. \quad (4.3)$$

Uniformly for $y = b + z$, $|z| \leq Z$,

$$\log Q_t(\max_j Y_j \leq y) = -N\Gamma(1 - \beta, y) + O_Z(e^{-b}). \quad (4.4)$$

Proof. The integrand in (4.2) is supported on $0 < s < Ny^{-\beta}$. The substitution $a = (N/s)^p$ gives

$$Ny^{-\beta}e^{-y} - \beta N \int_y^\infty e^{-a}a^{-\beta-1} da.$$

An integration by parts identifies this with $N \int_y^\infty e^{-a}a^{-\beta} da$. As a function of s , the original integrand is nonincreasing, takes limiting value e^{-y} at zero, and vanishes eventually. The elementary integral comparison for a nonincreasing nonnegative function therefore gives (4.3).

For contributing coordinates $a_j > y$,

$$0 \leq p_j(y) - q_j(y) \leq \frac{e^{-y}}{1 - e^{-y}} q_j(y).$$

When $y = b + z$ with z bounded, (4.2) and $Ne^{-b}b^{-\beta} = 1$ give $\sum q_j = O_Z(1)$, and hence $\Lambda_N(y) = N\Gamma(1 - \beta, y) + O_Z(e^{-b})$. Also $\max_j p_j(y) \leq e^{-y}/(1 - e^{-y}) = O_Z(e^{-b})$. The independent-product formula and $|\log(1 - u) + u| \leq u^2/[2(1 - u)]$ show that

$$\log Q_t(\max_j Y_j \leq y) = -\Lambda_N(y) + O\left(\sum_j p_j(y)^2\right).$$

The last sum is at most $\max_j p_j \cdot \Lambda_N = O_Z(e^{-b})$. □

4.2 Removing the small-sum constraint at logarithmic accuracy

Let $Q_t^{[y]} = Q_t(\cdot \mid \max_j Y_j \leq y)$. This is an independent product with caps $c_j = \min(a_j, y)$, since the event restricts each of only finitely many coordinates separately. Denote its total mean, variance, and density by μ_y, V_y, f_y . Define

$$D_y = \int_0^\infty e^{-h} f_y(\mu - h) dh. \quad (4.5)$$

Notice the *original* mean μ in this definition. Then

$$P_x(\max_j Y_j \leq y) = Q_t(\max_j Y_j \leq y) \frac{D_y}{D}. \quad (4.6)$$

Lemma 4.2 (Capping perturbs the normalization negligibly). *Uniformly for $y = b + z$, $|z| \leq Z$,*

$$\delta_y := \mu - \mu_y = O_Z(y), \quad |V - V_y| = O_Z(y^2), \quad (4.7)$$

and

$$\frac{D_y}{D} = 1 + O_{p,Z}(N^{-1/2} + y^2/N). \quad (4.8)$$

Proof. There are $K_y = \lfloor Ny^{-\beta} \rfloor$ possibly changed coordinates. For $a > y \geq 2$,

$$0 \leq m(a) - m(y) = \frac{y}{e^y - 1} - \frac{a}{e^a - 1} \leq Cye^{-y}, \quad |v(a) - v(y)| \leq Cy^2e^{-y}.$$

For the variance bound use (2.2) and the bound $a^2e^{-a} \leq y^2e^{-y}$ for $a \geq y \geq 2$, adjusting the absolute constant for the denominators. Since $K_y e^{-y} = O_Z(1)$, (4.7) follows. Thus $V_y \sim B_p N$.

Lemma 3.2, uniformly for the capped array, yields

$$D_y = \frac{1}{\sqrt{2\pi V_y}} \int_0^\infty e^{-h} \exp\left(-\frac{(\delta_y - h)^2}{2V_y}\right) dh + O_{p,Z}(N^{-1}).$$

The integral is $1 + O((\delta_y^2 + 1)/V_y)$, using $0 \leq 1 - e^{-u} \leq u$. Compare with (3.6) and use (4.7) to obtain (4.8). □

Proof of Theorem 1.1. Equations (4.6), (4.4), and (4.8) give

$$\log P_x(\max_j Y_j \leq b + z) = -N\Gamma(1 - \beta, b + z) + O_{p,Z}(e^{-b} + N^{-1/2} + b^2/N).$$

Because $e^{-b} = b^\beta/N$ and $b \sim \log N$, this proves (1.6).

The expansion has a particularly simple direct proof. Write $a = b + z + s$ in the gamma integral and use the defining equation for b :

$$N\Gamma(1 - \beta, b + z) = e^{-z} \int_0^\infty e^{-s} \left(1 + \frac{z + s}{b}\right)^{-\beta} ds. \quad (4.9)$$

Taylor's theorem, for fixed K and bounded z , gives an integrated remainder $O_{\beta,Z,K}(b^{-K-1})$. To check uniformity even for negative z , take $b > 2Z$: the Taylor segment stays at least $1/2$ away from the singularity at -1 , and the remainder is bounded by $Cb^{-K-1}(|z| + s)^{K+1}$, integrable against e^{-s} . The coefficient of b^{-r} is

$$\frac{(-1)^r (\beta)_r}{r!} \int_0^\infty e^{-s} (z + s)^r ds = (-1)^r (\beta)_r \sum_{k=0}^r \frac{z^k}{k!}.$$

This proves (1.7), without an interchange of an infinite asymptotic series and an integral. $\square$

Remark 4.3 (What the error does and does not mean). The $O(N^{-1/2})$ bound is deliberately conservative. It is adequate for every fixed inverse-logarithmic order because $N^{-1/2} = o(b^{-K})$ for every fixed K . It is not a claim of a sharp conditioning correction at the polynomial scale $1/N$. At moderate N the conditioning error can exceed several logarithmic terms; the numerical results in Section 8 illustrate this explicitly.

5 Poisson localization for polynomial weights

Proposition 5.1 (The tilted marked process). *Under Q_t , the process Ξ_t in (1.9) converges to Ξ . The same convergence holds under P_x , jointly with an independent $\text{Exp}(1)$ slack.*

Proof. Let g be a nonnegative continuous compactly supported function on $[0, \infty) \times \mathbb{R}$, with height support in $[L_0, L_1]$. Independence gives

$$\mathbb{E}_{Q_t} e^{-\langle g, \Xi_t \rangle} = \prod_j (1 - q_{j,t}), \quad q_{j,t} = \mathbb{E}_{Q_t} [1 - e^{-g(j/M, Y_j - b)}].$$

Only finitely many $q_{j,t}$ are nonzero. On their height support, $Y_j = b + z$ and $a_j = b(j/M)^{-p}$. Since $e^{-b} = 1/M$,

$$\begin{aligned} \sum_j q_{j,t} &= \frac{1}{M} \sum_j \int_{L_0}^{L_1} (1 - e^{-g(j/M, z)}) e^{-z} \\ &\quad \times \frac{\mathbf{1}_{b+z < b(j/M)^{-p}}}{1 - e^{-b(j/M)^{-p}}} dz. \end{aligned} \quad (5.1)$$

Where the indicator is nonzero, the denominator tends to one uniformly. For $u < 1$ the indicator tends to one; for $u > 1$ it tends to zero. Split the location integral into the two regions away from $u = 1$ and a strip of width 2δ around one. Ordinary Riemann sums give convergence on the first two regions, and a uniform bounded integrand controls the strip by $C\delta$. Thus

$$\sum_j q_{j,t} \longrightarrow \int_0^1 \int_{\mathbb{R}} (1 - e^{-g(u, z)}) e^{-z} dz du.$$

Also $\max_j q_{j,t} = O_g(M^{-1})$, so $\sum_j q_{j,t}^2 \rightarrow 0$. Taking logarithms proves convergence of Laplace functionals to that of Ξ . The same bounds on compact sets give local tightness, hence point-process convergence.

For the conditioned assertion, let $I_t = \{1, \dots, \lceil 2M \rceil\}$. Its tilted variance is at most $2M + 1 = o(N)$. For each fixed lower height L_0 , every coordinate outside I_t has cap at most $b/2^p < b + L_0$ for sufficiently large N . Therefore every point of Ξ_t above height L_0 is determined by I_t . Apply Corollary 3.4 and then the tilted convergence to obtain the joint Poisson and independent-slack limit. Exhausting the height axis by compact intervals completes the argument. $\square$

Corollary 5.2 (All fixed upper order statistics). *Order the contributions as $Y_{(1)} > Y_{(2)} > \dots$, with indices $J_{(1)}, J_{(2)}, \dots$. For each fixed r ,*

$$((J_{(k)}/M, Y_{(k)} - b))_{k=1}^r \Rightarrow ((U_k, -\log A_k))_{k=1}^r, \quad (5.2)$$

where $A_k = E_1 + \dots + E_k$, the E_i are independent $\text{Exp}(1)$, and the U_k are independent uniforms on $[0, 1]$, independent of all E_i . In particular

$$P_x(Y_{(r)} - b \leq z) \longrightarrow e^{-e^{-z}} \sum_{k=0}^{r-1} \frac{e^{-kz}}{k!}. \quad (5.3)$$

Proof. The intensity is a product of unit Lebesgue measure on $[0, 1]$ and $e^{-z}dz$. Under the substitution $v = e^{-z}$, the height intensity becomes Lebesgue measure on $(0, \infty)$, whose successive Poisson arrival times are A_k . Attaching independent uniform marks gives the displayed limiting process and hence the limiting order statistics.

To justify passage from vague convergence to top points, there is no escape in location above any fixed height: all such points lie in $[0, 2]$ for large N . Theorem 1.1 gives tightness of the highest height. The number above a low level L_0 tends to a Poisson variable with mean e^{-L_0} , which exceeds r with probability tending to one as $L_0 \rightarrow -\infty$. On a compact rectangle containing the top r points, the limiting process almost surely has no height ties or boundary points, so ordering is continuous. The formula (5.3) is the probability of at most $r - 1$ points above z . $\square$

Corollary 5.3 (No single summand carries the conditioned mass). *For polynomial weights,*

$$\frac{S}{x} \longrightarrow 1, \quad \frac{A_p N \max_j j^{-p} U_j}{b S} \longrightarrow 1 \quad \text{in } P_x\text{-probability.} \quad (5.4)$$

In particular the largest mass fraction tends to zero.

Proof. By Proposition 3.3, $T = \mu - H = \mu + O_{P_x}(1)$, while $\mu \sim A_p N$. Theorem 1.1 gives $\max_j Y_j = b + O_{P_x}(1)$. Divide these identities, using $b \rightarrow \infty$. $\square$

The extremes are concentrated in a vanishing fraction of the relevant indices, yet each contributes only order $\log N/N$ of the total mass. This distinction between location concentration and mass concentration is useful when interpreting the conditioned configurations.

6 The Gaussian bridge and joint independence

6.1 A variance-clock invariance principle

Lemma 6.1 (Tilted motion and conditioned bridge). *In both the polynomial and uniformly lacunary cases,*

$$\mathcal{B}_t \Rightarrow B \quad \text{under } Q_t, \quad (\mathcal{B}_t, H) \Rightarrow (B^\circ, E) \quad \text{under } P_x, \quad (6.1)$$

where B is standard Brownian motion and B°, E are independent.

Proof. Write $X_j = Y_j - m_j$. Since $\max_j v_j \leq 1$ and $V \asymp R$, the largest clock interval is $O(R^{-1})$. Lemma 2.1 gives

$$V^{-2} \sum_j \mathbb{E} X_j^4 \leq C V^{-2} \sum_j v_j^2 \leq C/V \rightarrow 0.$$

Taylor expansion of characteristic functions therefore proves the Gaussian finite-dimensional limits for disjoint partial-sum increments. Their limiting covariance in the exact clock is $\min(r, s)$. Interpolation at any fixed time changes the covariance by at most the largest clock interval and changes a random variable by $o_P(1)$.

For tightness, an increment between times $s < r$ is $V^{-1/2} \sum_j c_j X_j$, with $0 \leq c_j \leq 1$ and $\sum_j c_j v_j = V(r - s)$. Independence and the fourth-moment bound give

$$\begin{aligned} \mathbb{E}_{Q_t} |\mathcal{B}_t(r) - \mathcal{B}_t(s)|^4 &\leq C V^{-2} \left(\sum_j c_j^2 v_j \right)^2 \\ &\leq C(r - s)^2. \end{aligned}$$

This holds also for countably many summands by moment convergence. The fourth-moment tightness criterion on continuous paths follows, for example, by Markov's inequality on dyadic increments followed by the usual summable modulus-of-continuity bound. It proves the first convergence in (6.1).

Fix $0 < r_1 < \dots < r_d = c < 1$, and take the observed prefix I ending at the last clock knot before c . The clock mesh is negligible and $V_{I^c}/V \rightarrow 1 - c$. Include $Z_I/\sqrt{V}$ as the last coordinate in the observed Gaussian vector. Lemma 3.2, (3.8), and (3.6) show that, for bounded h , the joint likelihood is asymptotic to

$$e^{-h} g_c(Z_I/\sqrt{V}), \quad g_c(z) = \frac{1}{\sqrt{1-c}} \exp\left(-\frac{z^2}{2(1-c)}\right). \quad (6.2)$$

The approximation is uniform in $Z_I/\sqrt{V}$. The uniform bound $C_e e^{-h}$ permits integration over all $h \geq 0$. Weighting the Brownian finite-dimensional law by $g_c(B(c))$ gives the Brownian bridge law up to c : indeed g_c is the ratio of the normal density for completing $B(c)$ to zero in variance $1 - c$ to the unit-variance density at zero. Direct Gaussian conditioning gives covariance $\min(r, s) - rs$. The independent factor e^{-h} proves finite-dimensional independence of the slack.

For conditioned tightness on $[0, 1 - \varepsilon]$, use the bounded prefix likelihood in Corollary 3.4 and the tilted tightness. To control the terminal interval, take the tail after the last knot before $1 - \varepsilon$, with $0 < \varepsilon < 1/2$. Its variance is at most $\varepsilon V + 1$ and its complement has a fixed positive fraction of V . Its marginal likelihood is therefore bounded by a constant independent of ε . Kolmogorov's maximal inequality for reversed tail sums, first for finite tails and then by passage to the convergent series, gives

$$\limsup_{t \rightarrow \infty} P_x \left(\sup_{r \in [1-\varepsilon, 1]} |\mathcal{B}_t(r) - \mathcal{B}_t(1)| > \eta \right) \leq C\varepsilon/\eta^2. \quad (6.3)$$

One may replace reverse sums by a terminal sum minus forward partial sums to obtain the same bound. Interpolation cannot enlarge the maximum of the absolute endpoint values. Finally $\mathcal{B}_t(1) = -H/\sqrt{V} \rightarrow 0$ in probability. Letting $\varepsilon \downarrow 0$ proves tightness at the endpoint. The slack is tight by Proposition 3.3. The finite-dimensional limits identify the joint limit uniquely. $\square$

6.2 Why rare points are independent of Gaussian fluctuations

The following factorization makes the required independence explicit. Neither vanishing pairwise correlations nor separate marginal limits would suffice.

Lemma 6.2 (Rare-mark factorization). *Consider either tilted array and a marked process centered at c_t , where $c_t = b$ for polynomial weights and $c_t = \log n$ for lacunary weights. Suppose its compactly*

supported nonnegative test function g satisfies

$$q_j = \mathbb{E}[1 - e^{-g(\text{location}_j, Y_j - c_t)}], \quad \sum_j q_j = O_g(1).$$

For any bounded deterministic coefficients θ_j ,

$$\begin{aligned} & \mathbb{E} \left[\exp \left(i \sum_j \theta_j X_j / \sqrt{V} \right) e^{-\langle g, \Xi_t \rangle} \right] \\ & - \mathbb{E} \exp \left(i \sum_j \theta_j X_j / \sqrt{V} \right) \mathbb{E} e^{-\langle g, \Xi_t \rangle} \longrightarrow 0. \end{aligned} \quad (6.4)$$

The assertion also holds with the marks restricted to any deterministic coordinate subset.

Proof. Write $\phi_j = \mathbb{E} e^{i\theta_j X_j / \sqrt{V}}$. The j th factor in the first expectation is

$$\phi_j - q_j - e_j, \quad |e_j| \leq \frac{C}{\sqrt{V}} \mathbb{E}[|X_j|(1 - e^{-g(\text{location}_j, Y_j - c_t)})].$$

On the height support of g , $Y_j = c_t + O_g(1)$ and $0 \leq m_j \leq 1$, so $\sum_j |e_j| = O_g((c_t + 1)/\sqrt{V}) = o(1)$. Also $|\phi_j - 1| \leq C v_j / V \leq C/V$, by centering and $|e^{iu} - 1 - iu| \leq u^2/2$. Replacing the j th factor by $\phi_j(1 - q_j)$ has total error at most

$$\sum_j |e_j| + \sum_j q_j |\phi_j - 1| = o(1).$$

The telescoping product inequality is applicable since every exact expectation factor, as well as $\phi_j(1 - q_j)$, has modulus at most one. Independence factors the resulting product into the two expectations in (6.4). Finite products suffice for the marks; convergence of the centered sums handles an infinite Gaussian functional. $\square$

Taking θ_j to be the finite linear combination of partial-sum indicators corresponding to any chosen clock times proves joint independence of the tilted Brownian limit and the limiting marked Poisson process, whenever the latter has been established. Marginal tightness then gives joint tightness. The argument survives restriction of the marks to a prefix, which is crucial for conditioning.

6.3 Proof of the polynomial joint theorem

Proof of Theorem 1.2. Fix a compact height window for the marks and clock times $0 < r_1 < \dots < r_d < 1$. All marks in that window belong to the first $\lceil 2M \rceil$ coordinates for large N . The variance of that prefix is $o(V)$, so it lies before any fixed positive clock time. Choose $c \in (r_d, 1)$ and observe the entire coordinate prefix up to clock c , including its centered total $Z_I / \sqrt{V}$.

Under Q_t , Proposition 5.1, Lemma 6.1, and Lemma 6.2 give the joint limit of the observed Gaussian vector and the marks as independent Brownian motion and Ξ . Apply the exact conditional likelihood (3.8). As in (6.2), its limit is $e^{-h} g_c(B(c))$. This weight depends on the Gaussian component alone and on the separate slack variable, not on Ξ . It therefore changes Brownian motion into a Brownian bridge while leaving the Poisson process independent and unchanged, and produces an independent $\text{Exp}(1)$ slack.

This proves all finite-dimensional path, compactly supported mark, and slack limits jointly. Lemma 6.1, Proposition 5.1, and Proposition 3.3 give tightness of the three marginals, hence of their product. The resulting joint weak limit is exactly (1.10). The top-point continuity argument in Corollary 5.2 proves (1.11) and its stated independence properties. $\square$

7 Completion of the qualitative lacunary point-process question

For lacunary weights the extreme locations are spread over a positive fraction of the variance clock, unlike the polynomial case. The negligible-prefix argument alone would therefore not prove the independence theorem. Rare-mark factorization replaces it.

Proposition 7.1 (Lacunary marks under the tilted product). *Under (1.12),*

$$(\mathcal{B}_t, \Xi_t^{\text{lac}}) \Rightarrow (B, \Xi) \quad \text{under } Q_t, \quad (7.1)$$

with independence. The same assertion holds when marks are restricted to $j \leq \lfloor cn \rfloor$, with limiting intensity restricted to $0 \leq u \leq c$, for fixed $0 < c < 1$.

Proof. Fix a height window $[L_0, L_1]$. For $j \leq (1 - \varepsilon)n$, $a_j \geq q^{-\varepsilon n + O(1)}$, so its cap exceeds $\log n + L_1$ and its normalization $1 - e^{-a_j}$ tends uniformly to one. With location $u = j/n$ and height $Y_j - \log n = z$, the intensity calculation on this prefix is therefore the Riemann sum

$$\frac{1}{n} \sum_{j \leq (1-\varepsilon)n} \int (1 - e^{-g(j/n, z)}) e^{-z} dz.$$

Its limit is the required intensity integral over $[0, 1 - \varepsilon]$. For the remaining active indices, the probability of a contribution in this height window is at most

$$\frac{e^{-L_0}}{n(1 - e^{-\log n - L_0})}.$$

Their combined contribution is at most $C_g \varepsilon + O_g(n^{-1})$. Inactive coordinates have cap below one and cannot contribute for large n . Let $\varepsilon \downarrow 0$. The sum of the probabilities has the Poisson intensity limit and their maximum is $O_g(n^{-1})$, so the independent-product Laplace argument proves $\Xi_t^{\text{lac}} \Rightarrow \Xi$. Restricting to $j \leq cn$ proves the restricted statement.

Proposition 2.3 identifies the index and variance clocks up to $O_q(n^{-1})$. Lemma 6.1 and Lemma 6.2, now with $c_t = \log n$ and $V \sim n$, prove independent Brownian and Poisson limits. The moment bounds and the local count bounds give joint tightness. $\square$

Proof of Theorem 1.3. First retain only marks with $j \leq cn$, where $1/2 < c < 1$ is fixed. For finitely many clock times strictly below c , observe the whole prefix $I = \{1, \dots, \lfloor cn \rfloor\}$. By (2.12), its variance fraction tends to c . The tilted joint law of its Gaussian vector, its total, and its marks is described by Proposition 7.1. Applying (3.8) and Lemma 3.2 yields the limiting weight $e^{-h} g_c(B(c))$. Thus the marks on $[0, c]$ remain Poisson and independent, the Gaussian component becomes a bridge, and the slack becomes independent exponential.

It remains to remove the restriction on the marks, not to assume it harmless. Put $J = \{j > \lfloor cn \rfloor\}$. Its complement has at least a fixed positive fraction of the total variance, uniformly for $c \geq 1/2$, so its P_x marginal likelihood is bounded by a constant independent of c . Formula (4.1) and the fact that only the active indices contribute show, for each fixed L_0 ,

$$\limsup_{t \rightarrow \infty} P_x(\exists j > cn : Y_j - \log n > L_0) \leq C_q(1 - c)e^{-L_0}. \quad (7.2)$$

Indeed the corresponding tilted union bound has at most $(1 - c)n + 1$ terms, each bounded by $e^{-L_0}/[n(1 - e^{-\log n - L_0})]$; all inactive terms vanish. The limiting Poisson process has the analogous vanishing probability for a point in $(c, 1] \times (L_0, \infty)$.

For any fixed finite set of path times below one, choose c larger than all of them, use the truncated joint result, and then let $c \uparrow 1$ in (7.2). This gives the full joint finite-dimensional and compact-mark limits. The

path and slack components are tight by Lemma 6.1; the truncated-process approximation and (7.2) give tightness for the marks. Consequently the triple converges to the mutually independent triple in (1.13).

For completeness, the highest mark cannot escape to arbitrarily large height under P_x . Divide the active indices into two half-blocks. Each has a uniformly bounded marginal likelihood because its complement has asymptotic variance fraction 1/2. Under Q_t , the expected number above $\log n + L$ is at most Ce^{-L} , so the same upper bound, with another constant, holds under P_x . All marks above a fixed height have locations at most one. At low heights the limiting Poisson count grows without bound. The continuity argument for the top points therefore applies just as in Corollary 5.2, giving the uniform location and independent Gumbel height. $\square$

7.1 What changes between the two weight regimes

There is one universal limiting intensity but two different spatial normalizations. In the lacunary case an interior active coordinate has a cap much larger than $\log n$, and essentially all n active coordinates compete for the maximum. In the polynomial case the cap at $j = Nu$ is u^{-p} , a quantity of order one. To compete at height $b \sim \log N$, an index must satisfy $j \lesssim Nb^{-1/p}$. This moving cap boundary is why a naive centering by $\log N$ fails and why the location scale is M , not N .

The common bridge arises from the global mass constraint. The Poisson points survive independently because finitely many extreme contributions are only logarithmic in size, negligible on the square-root variance scale. Lemma 6.2 turns that statement into a joint characteristic-function calculation rather than an independence heuristic.

8 Reproducible symbolic and numerical diagnostics

The archive contains `code/verify.py`, the generated table `data/maximum_table.csv`, and a machine-readable execution record `data/verification.json`. The program performs six exact symbolic checks: the identity $v = m - am'$ and the five coefficient identities in (1.7) for orders zero through four. All six passed in the delivered run. It then evaluates the independent maximum distribution exactly as a finite mathematical product, and evaluates the conditional normalizers by numerical Fourier inversion with an analytically represented infinite tail.

All decimal evaluations are floating-point diagnostics, not interval certificates and not a substitute for the proofs. In particular, agreement under changes of numerical settings does not establish a bound on all rounding, truncation, or quadrature errors.

8.1 The exact formulas evaluated

For $y > 0$, let $K_y = \lfloor Ny^{-\beta} \rfloor$. The independent maximum formula is

$$\log Q_t(\max_j Y_j \leq y) = K_y \log(1 - e^{-y}) - \sum_{j=1}^{K_y} \log(1 - e^{-(N/j)^p}). \quad (8.1)$$

An equality case $a_j = y$ contributes zero and causes no ambiguity. No infinite truncation is needed in this formula.

If $\Phi_T(s) = \mathbb{E}_{Q_t} e^{is(T-\mu)}$, Fourier inversion and (3.4) give

$$D = \frac{1}{\pi} \int_0^\infty \Re \frac{\Phi_T(s)}{1 - is} ds. \quad (8.2)$$

The analogous D_y formula uses the capped law but remains centered at the original μ . Its characteristic function is the centered capped characteristic function multiplied by $e^{-is\delta_y}$. The density smoothness established in Lemma 3.2 justifies the inversion and the exponentially integrable slack kernel.

The code evaluates the first $J = \lceil 4N \rceil$ coordinates directly. For the rest, it uses the convergent expansion

$$\ell(a) := \log \frac{1 - e^{-a}}{a} = -\frac{a}{2} + \sum_{\substack{r \geq 2 \\ r \text{ even}}} c_r a^r, \quad c_r = \frac{B_r}{r r!}, \quad |a| < 2\pi. \quad (8.3)$$

Here B_r are the Bernoulli numbers. The centered characteristic logarithm for a tail coordinate is

$$\sum_{\substack{r \geq 2 \\ r \text{ even}}} c_r a^r ((1 - is)^r - 1 + irs).$$

Summation over $j > J$ uses the exact Hurwitz-zeta identity

$$\sum_{j > J} a_j^r = N^{pr} \zeta(pr, J + 1). \quad (8.4)$$

The linear term cancels on centering; the tail mean and variance are calculated from the same expansion. The delivered run retains the even terms through order 16 and integrates in the scaled variable $u = s\sqrt{V}$ up to $u = 30$. It does not represent discarded coordinates by zero, which would introduce an uncontrolled shift on the small-sum scale.

8.2 Results and finite-size effects

Table 1 reports $p = 2$, $z = 0$. The conditional column uses (4.6) and (8.2); the independent column uses (8.1). The incomplete-gamma column is $\exp[-N\Gamma(1/2, b)]$. Their common limiting Gumbel value is $e^{-1} = 0.3678794412\dots$

Table 1: Central maximum probabilities for polynomial weights j^{-2} .

N	b	Conditional P_x	Tilted Q_t	Gamma approximation
32	2.928492	0.450850950	0.415315863	0.414779547
128	4.141501	0.413874108	0.403560486	0.403369337
512	5.395538	0.399337502	0.396342140	0.396309268
2048	6.675404	0.392427364	0.391548647	0.391540557

The full CSV additionally contains $z = -1, 1$ at all four dimensions, for twelve parameter triples. At $N = 2048$ and $z = -1$, for example, the conditional probability is approximately 0.060955084, the independent probability 0.064524504, and the gamma approximation 0.064728174. The conditioning-to-tilt ratio is about 0.944681. Thus the table does not conceal a sizable lower-tail finite-size effect behind the limiting theorem. The constants implicit in uniform bounds may depend on the chosen bounded z -window.

An independent stability rerun at $p = 2$, $N = 128$, $z = 0$ changed J from $4N$ to $6N$, the tail order from 16 to 20, and the scaled integration cutoff from 30 to 36. The absolute change in the conditional CDF was 5.0×10^{-16} in the delivered environment. The mean and variance changed by approximately 2.8×10^{-14} and 1.4×10^{-14} , respectively. An independent high-precision integral evaluation of A_2 agreed with the gamma-zeta expression to about 2.6×10^{-25} . These are observations about numerical stability, not rigorous error bars on Table 1.

The executed environment was Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, SymPy 1.14.0, and mpmath 1.3.0. The script records the principal versions in its JSON output. Reproduction does not require GitHub access or downloading the repository.

9 Proof status, formalization, and limitations

9.1 A precise status ledger

Component	Status and scope
Uniform-series tilting and Gibbs conditioning	Classical framework; needed density and likelihood calculations rederived here. The general variance-fraction principle is already in [6].
Polynomial saddle constants and variance clock	Established background for the extension, proved here for self-containment; not advertised as new relative to the repository.
Incomplete-gamma conditional maximum law	Proved in Theorem 1.1; quantitative error for fixed $p > 1$ and bounded centered height.
Every fixed-order logarithmic correction	Proved with explicit coefficient polynomials and a remainder in (1.7); no growing-order uniformity claimed.
Polynomial marked extremes and localization	Proved in Theorem 1.2; spatial scale $N/b^{1/p}$ and all fixed upper order statistics.
Joint independence of bridge, points, and slack	Proved for polynomial and uniformly lacunary weights, with the displayed process topologies.
Lacunary Question 7	Its qualitative point-process and independence parts are answered. Its explicit joint-rate request is not answered.
Numerical work	Six symbolic identities passed; twelve floating-point CDF diagnostics and a stability rerun were executed. No interval or formal analytic certification of the decimal outputs.
Lean/Rocq verification	Not performed; no compiled formal theorem or repository build is claimed or included.
Global originality and peer review	Not established. Candidate contributions are identified relative to the inspected sources, not certified as historical firsts.

9.2 A realistic formalization path

The most economical first target is the maximum theorem rather than the complete point-process topology. The finite-product identity (8.1), the integral comparison (4.3), and the exact gamma integral (4.2) form a small, independently useful layer. These statements involve finite sums, a monotone real integral, and a single change of variables; they do not depend on the unformalized conditioning manuscripts.

The next layer should formalize the tilted truncated-exponential mean and variance, the product-capping identity (4.6), and a reusable density local limit theorem with explicit hypotheses. The proof boundary must state whether Fourier inversion, the normal characteristic function, and the moment tightness criterion are imported library theorems or independently proved. Merely defining an abstract density that is assumed to obey (3.1) would make the final maximum theorem conditional on that assumption.

The bridge and Poisson layer then requires probability kernels on a countable product, the likelihood disintegration (3.8), weak convergence of continuous paths, and Laplace-functional convergence for counting measures. The rare-mark factorization lemma is a useful finite-product starting point because its core estimate is a direct telescoping inequality. Numerical output should remain outside the trusted proof boundary unless a certified checker is supplied for the precise numerical assertion.

9.3 Limits of the statements

The constants in the polynomial estimates are not asserted uniform as $p \downarrow 1$ or $p \rightarrow \infty$. The weights are exactly j^{-p} for the gamma expansion; a slowly varying multiplier can alter its correction terms. The joint process convergence is qualitative, even though the scalar maximum law is quantitative. The point-process assertions use vague topology, not total variation between a discrete location lattice and a continuously located Poisson process; the latter comparison would be singular without coarsening or randomizing locations.

The proofs address the inequality $S \leq x$. A regular conditional law on $S = x$, a thin shell, or several simultaneous constraints requires its own likelihood and normalization. No such extension is silently included. Finally, the manuscript has not been externally refereed; the distinction between a complete argument written here and an independently checked theorem remains important.

10 Further research questions

1. Quantitative joint convergence. Complete the remaining rate component of the lacunary report's Question 7. Specify a metric combining path approximation with coarsened or bounded-Lipschitz point-process observables, and derive an explicit bound for joint independence. Exact total variation of continuous locations versus the deterministic index lattice is not an appropriate target. The rare-mark factorization proof already exposes an $O(\log R/\sqrt{R})$ cross-term before conditioning; controlling the complement likelihood and the final clock segment is the next obstacle.

2. The first genuine conditioning correction. Replace the conservative $O(N^{-1/2})$ error in Theorem 1.1 by an Edgeworth calculation for D_y/D , uniformly on a bounded height window. The exact quantities $\delta_y = \mu - \mu_y$ and $V - V_y$ identify the terms that must enter. Can one prove an explicit leading correction at order $(\log N)^2/N$, with a smaller controlled remainder? This is separate from the inverse-logarithmic series already proved.

3. Regularly varying weights. For decreasing $w_j = j^{-p}\ell(j)$ with slowly varying ℓ , seek a marked-extreme theorem in terms of the counting function $\nu_t(y) = \#\{j : tw_j > y\}$. The natural first centering equation is $e^{-b_t}\nu_t(b_t) \approx 1$. Determine the second-order regular variation assumptions that recover explicit corrections and distinguish them from the special exact gamma structure of pure powers.

4. Transition between polynomial and lacunary geometry. Study arrays in which the decay exponent or local log-slope changes with t . The limits $p \rightarrow \infty$ and $N \rightarrow \infty$ need not commute. Find a uniform description interpolating between localization in $o(N)$ indices and competition across a positive fraction of all active coordinates. The endpoint $p \downarrow 1$ has its own summability and scale issues and should be analyzed separately.

5. Nonuniform microscopic laws. Replace the uniforms by variables with density asymptotic to cu^{k-1} at zero. Their strongly tilted coordinates should be compared with gamma, rather than exponential, variables. Determine which regularity at zero and which global upper-tail assumptions suffice for joint Poisson-bridge independence. Even the logarithmic centering must account simultaneously for the gamma tail and the moving cap boundary; it should not be copied from (1.4).

6. Thin shells and multiple constraints. For $x - h_t \leq S \leq x$, determine how the limit depends on th_t , including a precise version of conditioning at $S = x$. With several weighted-sum constraints, formulate nondegeneracy hypotheses under which the Gaussian limit is projected off a finite-dimensional constraint

space. Identify when the extremes remain independent and when a constraint directly detects their locations.

7. Random weights and geometric phase. Establish quenched and annealed counterparts under random lacunarity or random regular variation. For geometric weights, combine the marked extremes with the phase-dependent microscopic boundary law already in [5, 6]. Joint phase mixing must be proved at the level of distributions; averaging a parameter inside a density is not generally the same as averaging the density itself.

8. Certified analytic computation. Turn (8.2)–(8.4) into an interval-certified algorithm with explicit Fourier-tail and Bernoulli-tail bounds, then determine bit complexity for a requested absolute or relative CDF error. The current numerical stability checks do not provide that certificate. A formal finite-product and gamma-integral layer would be a natural first contribution back to the repository.

11 Conclusion

The rare-event constraint has two mathematically distinct effects. It produces a Brownian bridge from collective square-root fluctuations and an exponential boundary slack, but it leaves logarithmic extremes asymptotically Poisson and independent of those objects. The locations of those extremes retain detailed information about the weight decay. They fill the active index range for uniformly lacunary weights and localize on $N/(\log N)^{1/p}$ indices for polynomial weights.

For pure polynomial weights, the entire fixed-order logarithmic maximum expansion is encoded by a single upper incomplete gamma integral. The key quantitative step is not the gamma asymptotics by itself, but the estimate showing that reimposing the small-sum constraint changes the logarithmic maximum probability by only $O(N^{-1/2})$ on bounded centered windows. The lacunary joint theorem answers a specifically identified qualitative research question in the inspected repository; the polynomial gamma law and localization provide a separate, more quantitative extension.

A Reproduction and source provenance

The archive is self-contained for TeX compilation. It contains the source `article.tex`, its compiled PDF, a `Makefile`, `README.md`, `CLAIM_STATUS.md`, the verification script, a requirements file, and generated numerical data. A standard TeX Live installation with the AMS packages, `newtx`, `microtype`, and `xurl` is sufficient. Run from the archive directory:

```
make pdf
python -m pip install -r requirements.txt
python code/verify.py
```

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The repository comparison was made at `a4268e78ebd0f6bf71ba609c07f4b3415748f532`. The two pinned manuscripts and the newer manuscript’s claim-status file were inspected; their earlier snapshot identifiers are provenance inside those manuscripts, not the snapshot used here. Narrow repository searches were used to locate related work. This is not an assertion that every file in the repository has been read or that all possible overlap has been excluded. The external comparison used the primary publisher records for [2, 3], Arias de Reyna’s paper, and the NIST special-function reference. The maximum, point-process, and joint-independence proofs above are supplied explicitly rather than inferred from those bibliographic records.

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- [4] NIST Digital Library of Mathematical Functions, Chapter 8, *Incomplete Gamma and Related Functions*, and Chapter 25, *Zeta and Related Functions*. Chapter 8; Section 25.5, integral representations. Consulted 30 September 2026. The specific gamma expansion used in this article is rederived in (4.9).
- [5] ProveIt research corpus, *Sharp Conditioning Laws for Fabius and Lacunary Random Series: A global simplex approximation, exact total-variation thresholds, and phase-dependent boundary layers*. Research manuscript prepared for V. Reshetnikov, 28 September 2026. Inspected at commit a4268e78ebd0f6bf71ba609c07f4b3415748f532. Pinned manuscript source. Question 7 is the qualitative point-process and independence target.
- [6] ProveIt research corpus, *Sharp Conditioning Laws for Uniform Random Series: Variance thresholds, boundary phases, Brownian bridges, and rare-event sampling*. Research manuscript prepared for V. Reshetnikov, 28 September 2026. Inspected at commit a4268e78ebd0f6bf71ba609c07f4b3415748f532. Pinned manuscript source; pinned claim-status ledger. The general variance criterion and polynomial clocks are existing repository results, not new claims of this article.